MAT 3007 – Optimization

Modeling and Linear Optimization

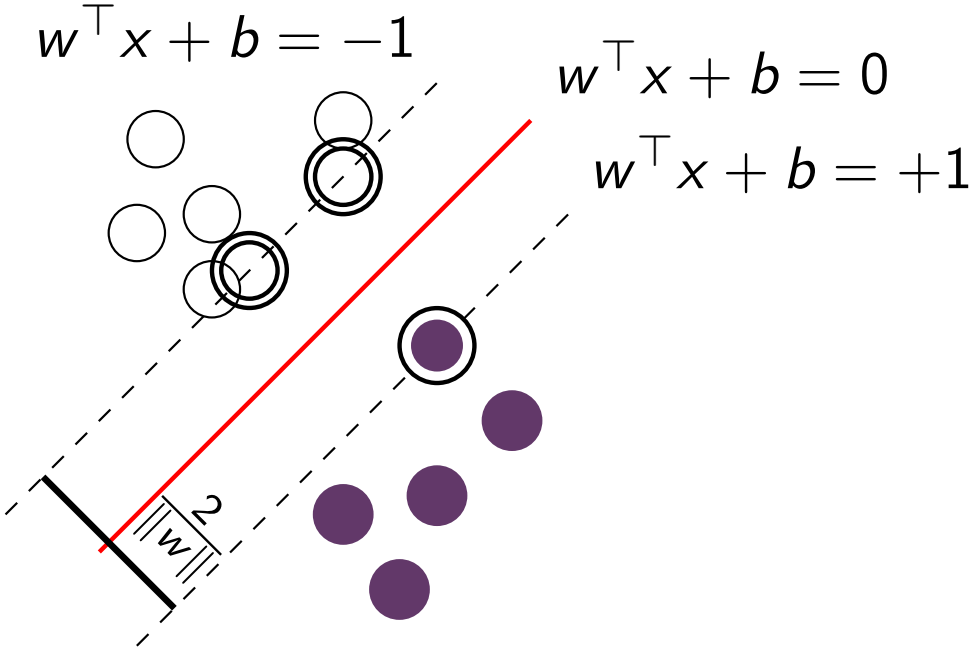
Lecture 04

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Recap

Recap: SVM



Hard-margin SVM for (linearly) separable data:

$$\min_{w,b} \quad \frac{1}{2}\|w\|^2 \quad \text{s.t.} \quad y_i(w^\top x_i + b) \geq 1, \quad i = 1, \ldots, m$$

Soft-margin SVM for potentially non-separable data:

$$\min_{w,b} \quad \frac{\lambda}{2}\|w\|^2 + \sum_{i=1}^{m} \max\{0, 1 - y_i(w^\top x_i + b)\}$$

Definition: Linear Program

Linear Problems

A linear optimization problem or linear program (LP) is an optimization problem in which the objective function and all constraint functions are linear/affine-linear in decision variables.

Example (Production Planning Problem):

$$\begin{aligned} \text{maximize} \quad & x_1 + 2x_2 \\ \text{subject to} \quad & x_1 \leq 100 \\ & 2x_2 \leq 200 \\ & x_1 + x_2 \leq 150 \\ & x_1, x_2 \geq 0. \end{aligned}$$

- Linear optimization is a subclass of continuous optimization.

General Formulation

A linear optimization problem can be generally written as:

$$
\begin{aligned}
\text{minimize/maximize}_{x\in\mathbb{R}^n} \quad & c^\top x & \\
\text{subject to} \quad & a_i^\top x \geq b_i & \forall\ i = 1, 2, \ldots, m_1 \\
& h_i^\top x \leq d_i & \forall\ i = 1, 2, \ldots, m_2 \\
& g_i^\top x = e_i & \forall\ i = 1, 2, \ldots, m_3 \\
& x_i \geq 0 & \forall\ i \in N_1 \\
& x_i \leq 0 & \forall\ i \in N_2 \\
& x_i \text{ free} & \forall\ i \in N_3
\end{aligned}
$$

$a_1^\top x \geq b_1$

$\vdots$

$a_{m_1}^\top x \geq b_{m_1}$

define $A = \begin{pmatrix} — a_1^\top — \\ — a_2^\top — \\ \vdots \\ — a_{m_1}^\top — \end{pmatrix}$

where $c, a_i, h_i, g_i \in \mathbb{R}^n$ and N_1, N_2, N_3 are subsets of $\{1, \ldots, n\}$.

$b = \begin{pmatrix} b_1 \\ b_2 \\ \vdots \\ b_{m_1} \end{pmatrix}$ $\quad Ax \geq b$ (element-wise)

Compact Formulation

We can write LPs in a more compact way:

$$
\begin{aligned}
\text{minimize/maximize}_{x\in\mathbb{R}^n} \quad & c^\top x \\
\text{subject to} \quad & Ax \geq b \\
& Hx \leq d \\
& Gx = e \\
& x_i \geq 0 \quad \forall\ i \in N_1 \\
& x_i \leq 0 \quad \forall\ i \in N_2 \\
& x_i \text{ free} \quad \forall\ i \in N_3
\end{aligned}
$$

- A, H, and G are matrices (with dimensions $m_1 \times n$, $m_2 \times n$ and $m_3 \times n$).
- b, d, and e are vectors (with dim. $m_1 \times 1$, $m_2 \times 1$ and $m_3 \times 1$).

Standard Form of LPs

In order to study LPs more systematically, we want to have a standard form for LPs:

$$\begin{aligned} \text{minimize}_{x\in\mathbb{R}^n} \quad & c^\top x \\ \text{subject to} \quad & Ax = b \\ & x \geq 0 \end{aligned} \tag{1}$$

where $x \in \mathbb{R}^n$, A is an $m \times n$ matrix and $b \in \mathbb{R}^m$.

Remark:

- The definition of "standard form" may differ from book to book. We use (1) as the standard form in this course (consistant with Bertsimas and Tsitsiklis's book).

Transformation to Standard Forms

1. If the objective is maximization:
   - replace c with $-c$ and change it to minimization.
2. If there is inequality constraints $Ax \leq b$ or $Ax \geq b$:
   - add decision variable s, also called slack variable.
   - replace the inequality with $Ax + s = b, s \geq 0$ or $Ax - s = b, s \geq 0$.
3. If there is non-positivity constraint $x_i \leq 0$:
   - replace all x_i wtih $-y_i$, where $y_i \geq 0$.
4. If there is "free" variables x_i (no constraints on x_i):
   - replace x_i with $x_i^+ - x_i^-$, where $x_i^+ \geq 0$, $x_i^- \geq 0$

Example: Production Planning

why not standard form?

① max → min

② ineq. constraints → add slack variable

$$
\begin{array}{lrlll}
\text{maximize} & x_1 & +2x_2 & & \\
\text{subject to} & x_1 & & \leq 100 & \\
 & & 2x_2 & \leq 200 & \\
 & x_1 & +x_2 & \leq 150 & \\
 & x_1, & x_2 & \geq 0 &
\end{array}
$$

Standard Form:

$$
\begin{array}{lrlllll}
\text{minimize} & -x_1 & -2x_2 & & & & \\
\text{subject to} & x_1 & & +s_1 & & & = 100 \\
 & & 2x_2 & & +s_2 & & = 200 \\
 & x_1 & +x_2 & & & +s_3 & = 150 \\
 & x_1, & x_2, & s_1, & s_2, & s_3 & \geq 0
\end{array}
$$

$$A = \begin{pmatrix} 1 & 0 & 1 & 0 & 0 \\ 0 & 2 & 0 & 1 & 0 \\ 1 & 1 & 0 & 0 & 1 \end{pmatrix}$$

$$Ax = \begin{pmatrix} 100 \\ 200 \\ 150 \end{pmatrix} =: b$$

Example: Support Vector Machines

why not standard form?

① ineq. constraint → slack.

② w, b are free

$$
\begin{aligned}
\text{minimize}_{w,b,t} \quad & \sum_i t_i \\
\text{subject to} \quad & y_i(x_i^\top w + b) + t_i \geq 1, \quad \forall\, i \\
& t_i \geq 0 \qquad\qquad\quad \forall\, i.
\end{aligned}
$$

$w = w^+ - w^-$

$b = b^+ - b^-$

↑ vectors

- Replace $w = w^+ - w^-$, $b = b^+ - b^-$, where w^+, w^-, $b^+, b^- \geq 0$.
- Add slack variables to eliminate inequality constraints.

Standard Form for SVMs

$$
\begin{aligned}
\min_{w^+,w^-,b^+,b^-,t,s} \quad & \sum_i t_i \\
\text{subject to} \quad & y_i(x_i^\top w^+ - x_i^\top w^- + b^+ - b^-) + t_i - s_i = 1 \quad \forall i \\
& w^+, w^-, b^+, b^- \geq 0 \\
& t_i, s_i \geq 0 \qquad \forall\ i.
\end{aligned}
$$

Standard Form: Discussion

- Standard form is mainly used for analysis purposes. We do not need to write a problem in standard form unless necessary.
- Usually we just represent the problem in a way that makes it easy to understand.
- Transforming an LP into the standard form is an important skill. It is helpful for analyzing LP problems as well as when using software to solve it.

Linear Optimization: Implementation

Using Software to Solve LPs

In this course, we will mainly work with MATLAB & Python to solve LPs and other optimization problems:

- Download a package called CVX / CVXPY.
- Read the instruction documents.
- `http://cvxr.com` or `https://cvxpy.org`.
- More guidelines on BB and in the tutorials.

Example: Production Planning Problem

$$
\begin{array}{lrrl}
\text{maximize} & x_1 & +2x_2 & \\
\text{subject to} & x_1 & & \leq 100 \\
& & 2x_2 & \leq 200 \\
& x_1 & +x_2 & \leq 150 \\
& x_1, & x_2 & \geq 0.
\end{array}
$$

Code: Production Plan (MATLAB)

```
% Production Planning Problem

cvx_begin quiet
variables x1  x2;
     maximize 1 * x1 + 2 * x2
     subject to
          x1          <= 100;
               2 * x2 <= 200;
          x1 +     x2 <= 150;
          [x1, x2] >= 0;
cvx_end

fprintf(1,'Optimal solution: [x1,x2] = [%1.2f,%1.2f]; Optimal value: %g.\n'
     ,x1,x2,cvx_optval);
```

Code: Production Plan (Python)

```
1  import numpy as np
2  import cvxpy as cp
3
4  c = np.array([1,2])
5  A = np.array([[1,0],[0,2],[1,1]])
6  b = np.array([100,200,150])
7
8  x = cp.Variable(2)
9  objective = cp.Maximize(c@x)
10 constraints = [A@x <= b, x >= 0]
11 prob = cp.Problem(objective, constraints)
12
13 prob.solve()
14 x.value
15
16 print(x.value)
17 print(prob.value)
```

Example: Shortest Path Problem

Let $G = (V, E)$ be a graph where $V = \{1, ..., n\}$ is the set of nodes and E is the set of edges.

- We denote the source node by 1 and the terminal node by n.
- We use w_{ij} to denote the distance from i to j. In general, w_{ij} does not necessarily equal w_{ji} (i.e., a directed graph).
- We assume E contains all pairs of (directed) nodes: If there was no edge for (i, j), we can just set w_{ij} extremely large (e.g., larger than n times the maximum of the rest of w_{ij}).

We want to write a general shortest path solver using LPs:

- Input: A weight matrix $W = \{w_{ij}\}_{i,j=1,...,n}$.
- Output: The shortest path from 1 to n and its distance.

Example: Shortest Path Problem

We have derived the optimization formulation for this problem:

$$
\begin{aligned}
\text{minimize} \quad & \textstyle\sum_{(i,j)\in E} w_{ij} x_{ij} \\
\text{subject to} \quad & \textstyle\sum_j x_{1j} = 1 \\
& \textstyle\sum_j x_{jn} = 1 \\
& \textstyle\sum_j x_{ij} = \sum_j x_{ji}, \quad \forall\ i \neq 1, n \\
& x_{ij} \in \{0,1\}, \qquad \forall\ (i,j) \in E.
\end{aligned}
$$

path length

start from node 1

end at node n.

edges form a path

Example: Shortest Path Problem

For simplicity of implementation, we further include x_{ii} as decision variables and set w_{ii} to be very large.

Decision variables: a matrix $X = \{x_{ij}\}_{i,j=1,\ldots,n}$

Objective function: $\sum_{i,j} w_{ij}x_{ij}$

- MATLAB representation: `sum(sum(W.*X))`.

Constraints:

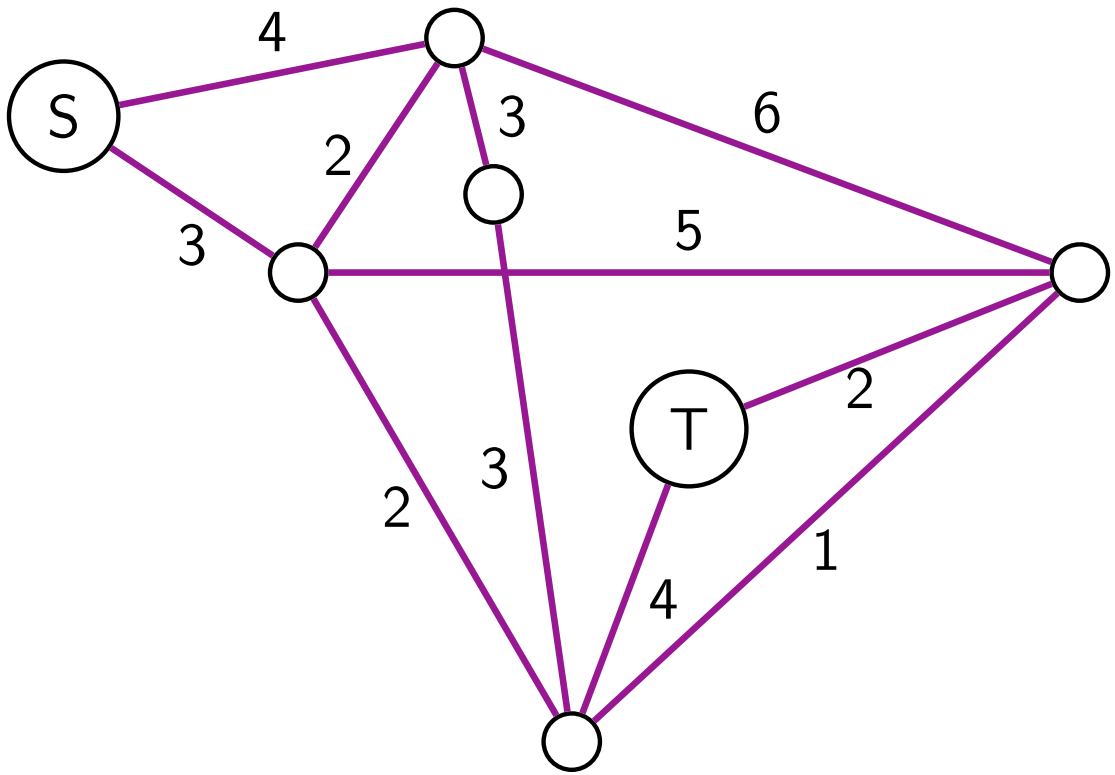
- `sum(X(1,:)) = 1`.
- `sum(X(:,n)) = 1`.
- `sum(X(i,:)) - sum(X(:,i)) = 0` for $i \neq 1, n$.

Integer constraints:

- We relax $x_{ij} \in \{0, 1\}$ to $0 \leq x_{ij} \leq 1$. For this problem, this will not change the solution ($\rightsquigarrow$ more later).

Example: Shortest Path Problem

After discussing the general setup, we can solve a specific problem:



Code: Shortest Path (MATLAB)

```
%Input the weight matrix W matrix
M = 100;
W = [M, 3, 4, M, M, M, M;
    3, M, 2, M, 2, 5, M;
    4, 2, M, 3, M, 6, M;
    M, M, 3, M, 3, 2, M;
    M, 2, M, 3, M, 1, 4;
    M, 5, 6, 2, 1, M, 2;
    M, M, M, M, 4, 2, M];

[n, ~] = size(W);

cvx_begin quiet
variables x(n,n)
    minimize sum(sum(W .* x))
    subject to
        sum(x(1,:)) == 1;
        sum(x(:,n)) == 1;
        sum(x(2:n-1,:), 2) - sum(x(:,2:n-1))' == 0;
        0 <= x <= 1;
cvx_end

fprintf(1,'Optimal value: %g.\n',cvx_optval);
```

(Re-)Formulating and Modeling Linear Problems

Reformulation Principles

We have already seen how the support vector machine problem can be reformulated as a linear program.

Core Techniques:

- Introduce auxiliary variables.
- Relax binding constraints ("$=$" $\rightarrow$ "$\leq$").

$\rightsquigarrow$ Minimization of the objective function "pushes" the relaxed variables back to the original constraint.

- Utilize equivalent representations.

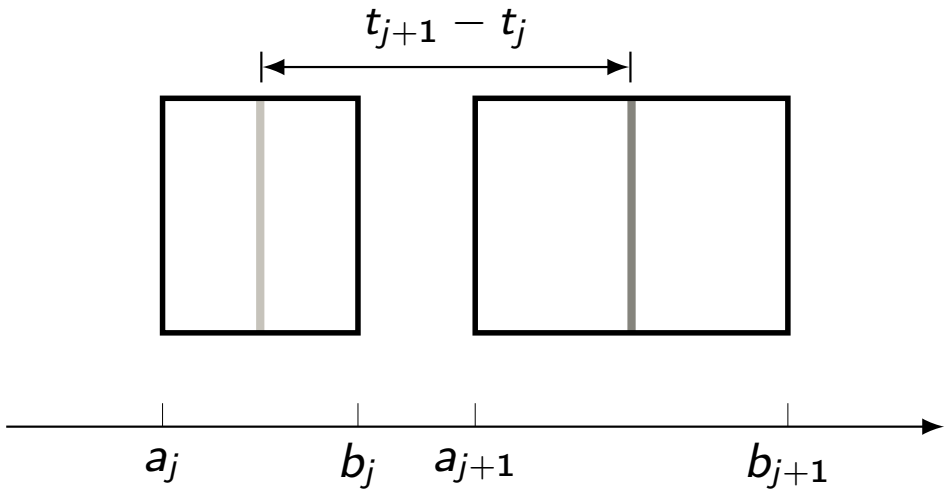
We now discuss several more examples: minimax problems, absolute values, ...

Modeling: Minimax Problems

Air Traffic Control Problem

An air traffic controller needs to control the landing times of n aircrafts:

- Flights must land in the order $1, \ldots, n$.
- Flight j must land in time interval $[a_j, b_j]$.
- The objective is to maximize the minimum separation time, which is the interval between two landings.



An Optimization Formulation

Decision variables:

- Let t_j be the landing time of the flight j.

Optimization problem:

$$\begin{aligned} \max \quad & \min_{j=1,\ldots,n-1}\{t_{j+1} - t_j\} \\ \text{s.t.} \quad & a_j \le t_j \le b_j, && j = 1, \ldots, n \\ & t_j \le t_{j+1}, && j = 1, \ldots, n-1. \end{aligned}$$

Observation:

- The objective function is not a linear function. We call it a maximin objective.

Reformulation as LP

Introduce an auxiliary variable Δ that satisfies:

$$\Delta \leq \min_{j=1,\ldots,n-1} \{t_{j+1} - t_j\}.$$

Later we will show the original problem is equivalent to:

$$\begin{aligned} \text{maximize}_{\Delta,t} \quad & \Delta \\ \text{subject to} \quad & \Delta \leq \min_{j=1,\ldots,n-1}\{t_{j+1} - t_j\} \\ & a_j \leq t_j \leq b_j, \qquad j = 1, \ldots, n \\ & t_j \leq t_{j+1}, \qquad j = 1, \ldots, n-1. \end{aligned}$$

Reformulation as LP

Since $\Delta \leq \min_{j=1,\ldots,n-1}\{t_{j+1} - t_j\}$ is equivalent to $\Delta \leq t_{j+1} - t_j$ for all j, we can rewrit the problem as

$$\begin{aligned}
\text{maximize}_{\Delta,t} \quad & \Delta \\
\text{subject to} \quad & t_{j+1} - t_j - \Delta \geq 0, \quad j = 1, \ldots, n-1 \\
& a_j \leq t_j \leq b_j, \quad j = 1, \ldots, n \\
& t_j \leq t_{j+1}, \quad j = 1, \ldots, n-1.
\end{aligned}$$

- It is an LP problem now.

Minimax Objective

Similar to the air traffic control problem, we are also interested in a minimax objective:

$$\begin{array}{ll} \text{minimize}_x & \max_{i=1,\ldots,n}\{c_i^\top x + d_i\} \\ \text{subject to} & Ax = b \\ & x \geq 0. \end{array}$$

We can deal with it in a similar manner:

- Introduce y satisfies $y \geq \max_{i=1,\ldots,n}\{c_i^\top x + d_i\}$:

$$\begin{array}{ll} \text{minimize}_{x,y} & y \\ \text{subject to} & y \geq \max_{i=1,\ldots,n}\{c_i^\top x + d_i\} \\ & Ax = b \\ & x \geq 0. \end{array}$$

- Then rewrite $y \geq \max_{i=1,\ldots,n}\{c_i^\top x + d_i\}$ as $y \geq c_i^\top x + d_i,\ \forall i$.

Modeling: Absolute Values

Dealing with Absolute Values

Problems with absolute values might be handled as well by LPs:

$$\begin{aligned} \text{minimize} \quad & \sum_{i=1}^n |x_i| \\ \text{s.t.} \quad & Ax = b. \end{aligned}$$

introduce t_i and

$t_i \geq |x_i| \quad \forall i$

$\Leftrightarrow$ $t_i \geq x_i$, $t_i \geq -x_i$

This can be equivalently written as

$$\begin{aligned} \text{minimize}_{x,t} \quad & \sum_{i=1}^n t_i \\ \text{s.t.} \quad & t_i \geq x_i \\ & t_i \geq -x_i \\ & Ax = b. \end{aligned}$$

- Similar ideas can be applied for constraints like $|a^\top x + b| \leq c$.

A General Modeling Tool

Consider the optimization problems:

$$\text{minimize}_x \ \sum_{i=1}^n f_i(x) \quad \text{s.t.} \quad x \in \Omega \tag{2}$$

$$\text{minimize}_{x,t} \ \sum_{i=1}^n t_i \quad \text{s.t.} \quad x \in \Omega, \quad f_i(x) \le t_i, \quad \forall\, i, \tag{3}$$

where $f_i : \mathbb{R}^n \to \mathbb{R}$ are given and $\Omega \subset \mathbb{R}^n$ is the feasible set.

Lemma: Modeling Tool

The problems (2) and (3) are equivalent in the following way:

- If x^* is a sol. of (2), then $(x^*, t_1^*, \cdots, t_n^*)$ with $t_i^* = f_i(x^*)$ is an optimal solution of (3).
- If (x^*, t^*) is an optimal solution of (3), then x^* is an opt. sol. of (2).

Also, both problems have the same optimal value.

proof (sketch): We assume $n=1$ for simplicity.

<1> If $x^* \in \Omega$ is an optimal sol. to (2), then (x^*, t^*) with $t^* = f(x^*)$ is also a feasible sol. to (3). Also, by optimality of x^* in (2), we have

$$f(x) \geq f(x^*), \quad \forall x \in \Omega.$$

Then consider any feasible point (x,t) of problem (3), we have

$$t \geq f(x) \geq f(x^*) = t^* \quad \Rightarrow (x^*, t^*) \text{ is optimal for (3)}$$

<2> If (x^*, t^*) is an optimal sol. to (3), then by feasibility, $x^* \in \Omega$, so x^* is also feasible for (2).

Also, observe that for any $x \in \Omega$, (x,t) with $t = f(x)$ is a feasible solution to (3). Therefore, by optimality of (x^*, t^*) for (3), we have $f(x) = t \geq t^*$. Also, because (x^*, t^*) is feasible for (3), we have $t^* \geq f(x^*)$. We conclude that $f(x) = t \geq t^* \geq f(x^*)$, $\forall x \in \Omega$.

$\Rightarrow x^*$ is optimal for (2). □

Absolute Values: An Observation

when we have a maximization problem, we use $t_i \le f_i(x)$

Consider a similar problem:

$$\begin{aligned} \text{maximize} \quad & \sum_{i=1}^n |x_i| \\ \text{s.t.} \quad & Ax = b. \end{aligned}$$

Can we use the similar idea and transform it into:

$$\begin{aligned} \text{maximize} \quad & \sum_{i=1}^n t_i \\ \text{s.t.} \quad & t_i \geq x_i \\ & t_i \geq -x_i \\ & Ax = b. \end{aligned}$$

- Answer:

Modeling: Fractional Programming

Consider the problem:

$$\begin{aligned} \text{minimize}_x \quad & \frac{c^\top x + d}{e^\top x + f} \\ \text{s.t.} \quad & Ax \leq b. \end{aligned} \tag{4}$$

- We assume that $e^\top x + f > 0$ for any x satisfying $Ax \leq b$.
- Production Planning: The objective of the company might be based on maximizing the ratio: (*total profit*)/(*total production costs*).

Any idea to transform it to LP?

- Define:

$$y = \frac{x}{e^\top x + f}, \qquad z = \frac{1}{e^\top x + f}.$$

LFP: A Reformulation

We can write the problem as:

$$\begin{aligned} \text{minimize}_{y,z} \quad & c^\top y + dz \\ \text{s.t.} \quad & Ay - bz \leq 0 \\ & e^\top y + fz = 1 \\ & z \geq 0 \end{aligned} \tag{5}$$

- This is an LP.
- Why are they equivalent? (in optimal value)

$\rightsquigarrow$ See Boyd and Vandenberghe for details (supplemental reading: page 151).

LFP: A Reformulation

proof idea: <1> for any feasible point x of (4), we construct

$$y \triangleq \frac{x}{e^T x + f} \text{ (vector)}, \quad z = \frac{1}{e^T x + f} \text{ (scalar)}$$

Try to show that (y, z) is feasible for (5), and the objective value of x in (4) is the same as objective value of (y, z) in (5).

$\Rightarrow$ optimal value of (5) $\leq$ optimal value of (4)

<2> For any feasible point (y, z) of (5)

① if $z \neq 0$, we construct $x = \frac{y}{z}$. Try to show x is feasible for (4) and have the same objective value.

② if $z = 0$?

Recap

Up to now, we have learned how to:

- Formulate linear optimization problems.
- Transform an LP into a standard form.
- Use MATLAB / Python to solve linear optimization problems.

Next:

- How to solve linear optimization problem?
- We will start with some basic properties of LPs.

Questions?